

Shaping the future of Bluetooth audio

In the context of Bluetooth audio, the pioneering strides made by Qualcomm Technologies, Inc. are truly captivating. Their consistent trailblazing was prominently displayed in 2023 with the introduction of Auracast broadcast audio. This groundbreaking contribution not only turned industry heads but also ignited a fervour among end-users, heralding a new era in audio experiences.

The Auracast Experience event, held at Mobile World Congress and other global industry gatherings, allowed users to immerse themselves in the capabilities of Auracast. For the first time, multiple Bluetooth wireless audio broadcasts coexisted in the same location, demonstrating the versatility and potential of this technology.

Auracast, fuelled by earbuds harnessed with the Qualcomm S5 Sound Platform, has opened uncharted territories for audio-sharing experiences. These earbuds serve as a prime example of the innovative implementation of Bluetooth SoCs (systems on chips) tailored specifically to support Auracast broadcast audio.

Beyond its prowess in audio sharing, Auracast has successfully tackled a significant challenge in Bluetooth audio—latency. In the gaming realm, where latency is paramount, Qualcomm Technologies has, in just two years, transformed Bluetooth audio latency from 200ms to a mere 20ms. This monumental achievement, facilitated by the Snapdragon Sound Technology Suite and Bluetooth LE Audio, has revolutionised gaming and spatial audio experiences.

According to Qualcomm's State of Sound consumer research, a noteworthy percentage of users struggle to hear in busy environments,

Simple upgrades, new experiences!

Numerous innovative applications and products are on the horizon, promising transformative experiences. Key use cases span from receiving essential flight information at airports to enjoying public TV audio and facilitating seamless music sharing among friends. Additionally, Auracast holds great potential for assisted listening, particularly for individuals with hearing devices. Unlocking the Auracast experience:

- Both transmitter and receiver sides need to be LE Audio compliant.
- Modern smartphones can be easily upgraded for transmission through a software update.
- Older devices can enhance compatibility with an Auracast bridge dongle.
- Initial consumer products supporting Auracast broadcast audio on the receiving end are expected by year-end.

To harness the rapid growth in the market, developing and deploying a turnkey platform are crucial to meet the aggressive demands of time-to-market.

presenting an opportune space for Auracast to address individual hearing needs.

Looking ahead, 2024 promises substantial growth in health and wellness monitoring using Bluetooth technology. The combination of Bluetooth LE Audio's lower power and flexibility, coupled with the advanced AI features in Qualcomm's latest S7 Sound Platforms, foretells seamless and responsive user experiences with potential positive impacts on daily lives.

Auracast's solution for TWS earbuds

The limitations of the traditional Bluetooth 'Classic' standard become evident, especially with the surging popularity of True Wireless Stereo (TWS) earbuds. Classic Bluetooth's lack of native TWS support led to the adoption of proprietary technologies, introducing latency and power consumption issues. Auracast steps in to overcome these challenges, offering a seamless and advanced audio solution.

Auracast draws its strength from the LE Audio specification, built upon Bluetooth LE. This cutting-edge technology introduces dual-stream audio, allowing both earbuds direct connectivity to the audio source. The result is a reduction in power consumption and latency,

unlocking the full potential of TWS earbuds.

As Bluetooth LE is projected to dominate the market by 2026, accounting for 43% of total shipments, Auracast takes centre stage in headsets, hearing aids, and hearables. The coexistence of Bluetooth dual mode ensures backward compatibility, highlighting Auracast as the forefront of Bluetooth audio evolution.

The LE Audio specification introduces key advancements that Auracast maximises—from lower power consumption to synchronised multichannel audio and support for one-to-many audio broadcasting. Thanks to the visionary contribution of CEVA, Auracast has seamlessly integrated these innovations, bringing a new dimension to audio experiences.

To enable the broadcast capability of Auracast, a sophisticated additional layer, the generic audio framework, sits atop the Bluetooth LE protocol stack. This framework, a collection of services and profiles, is the magic behind the immersive broadcast audio experience Auracast offers.

What sets Auracast apart is its user-centric design. The transmitter device operates without the need-to-know which receiver devices are connected. Accessing Auracast is made simple for consumers, with a user-friendly menu, possibly right on your phone's home screen.